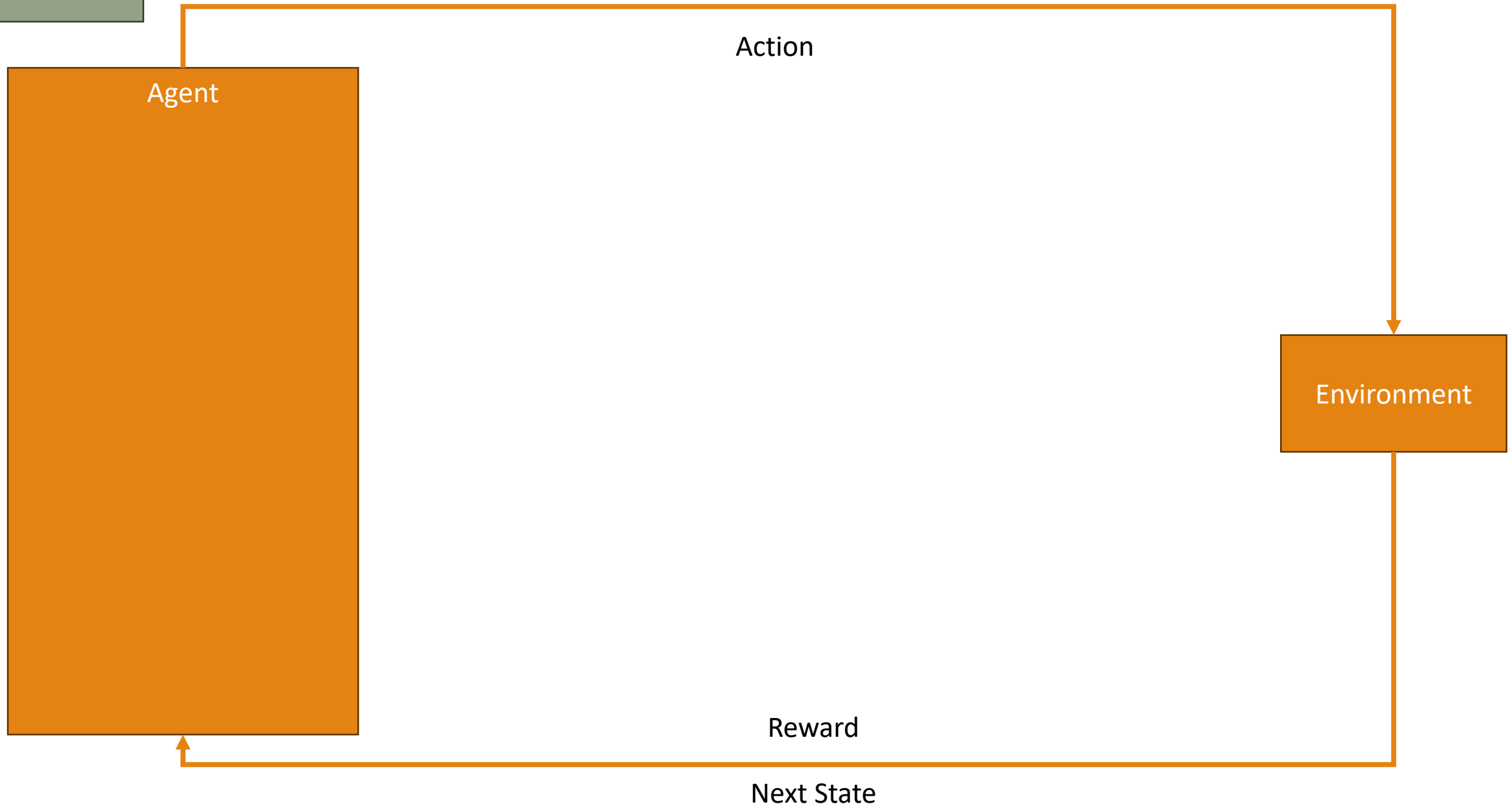
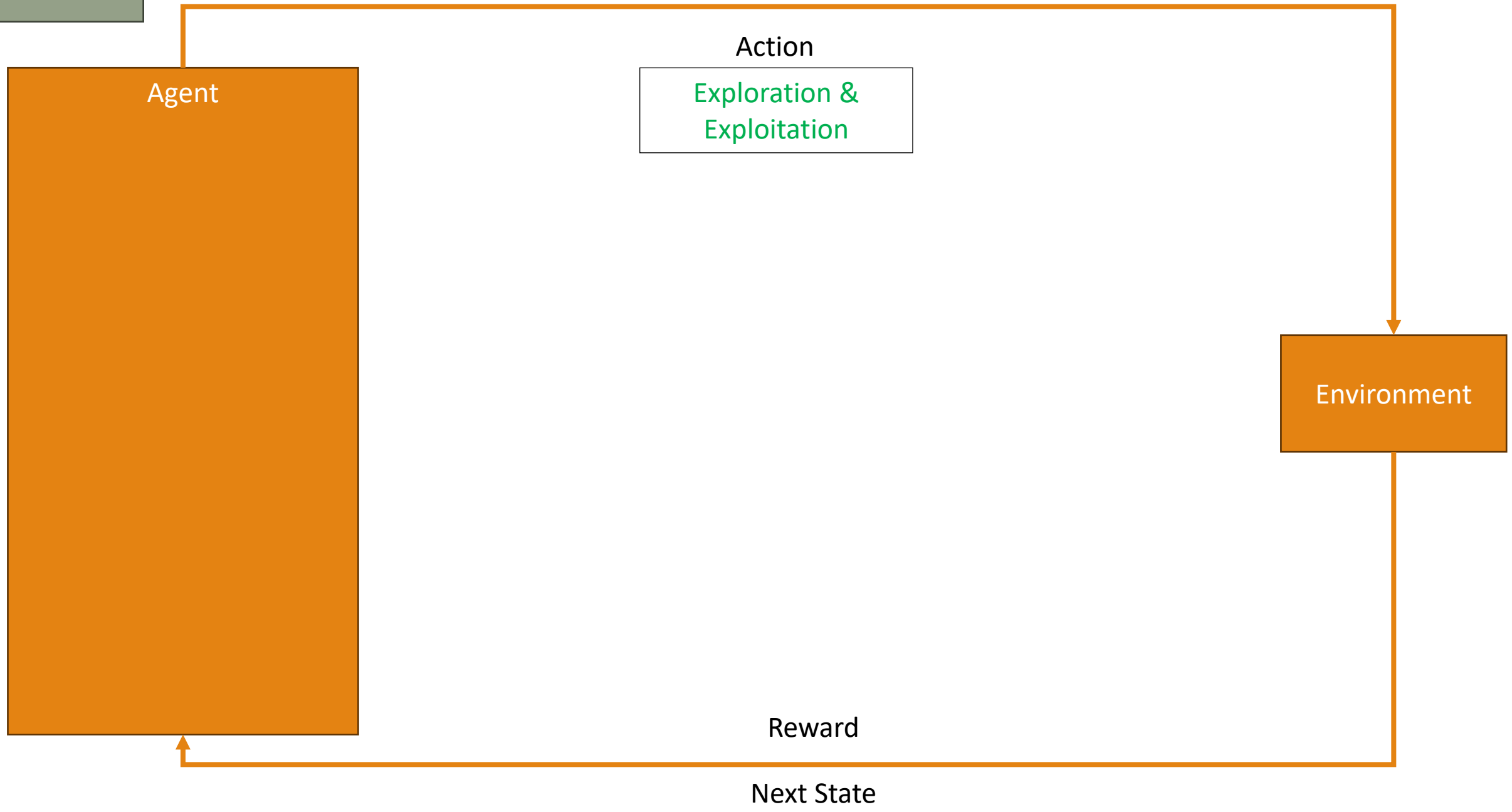
Model-free Prediction

Blink Sakulkueakulsuk

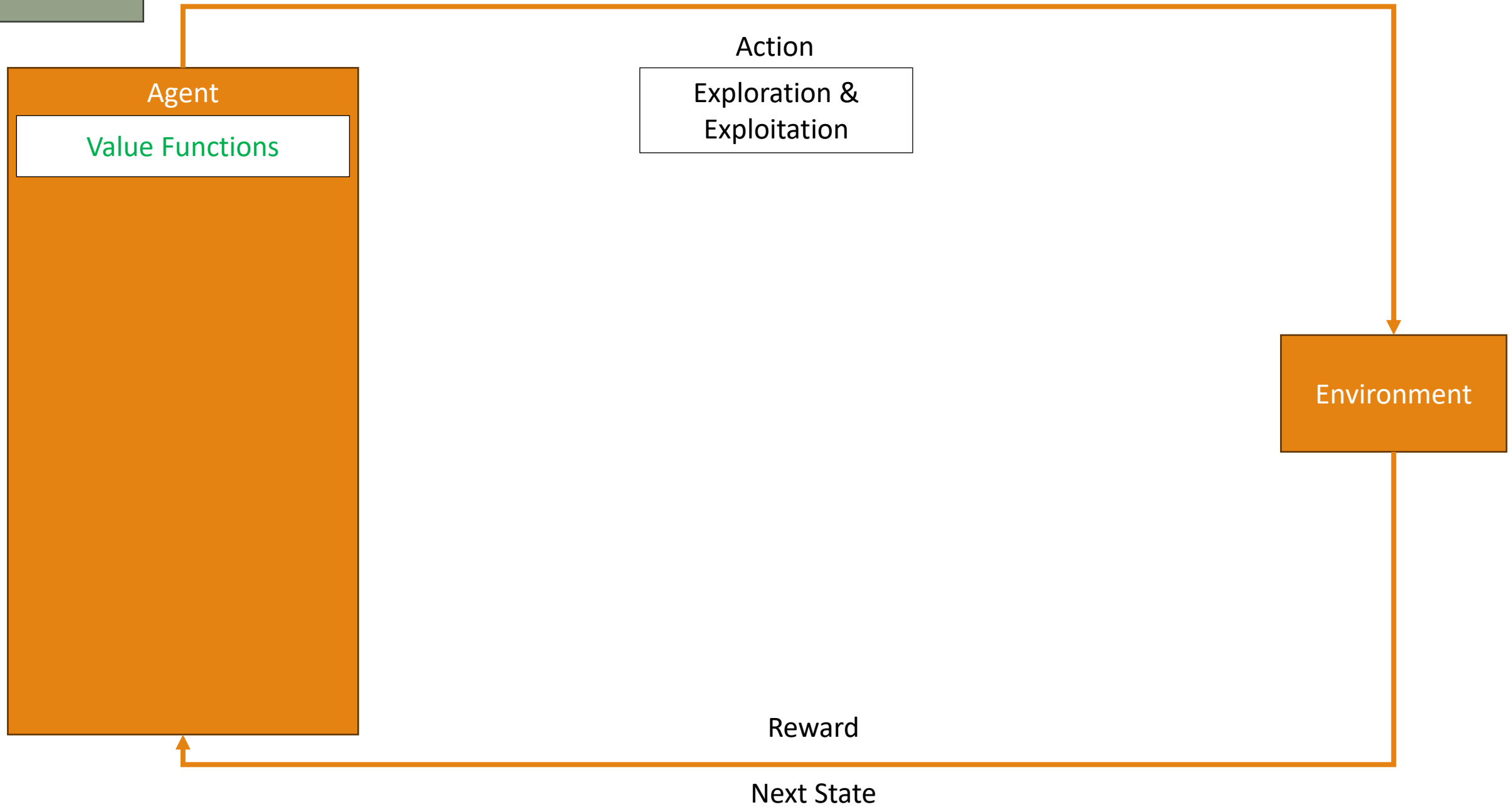
Lecture 1



Lecture 2


Action
Exploration &
Exploitation
Agent
Environment
Reward
Next State

Lecture 3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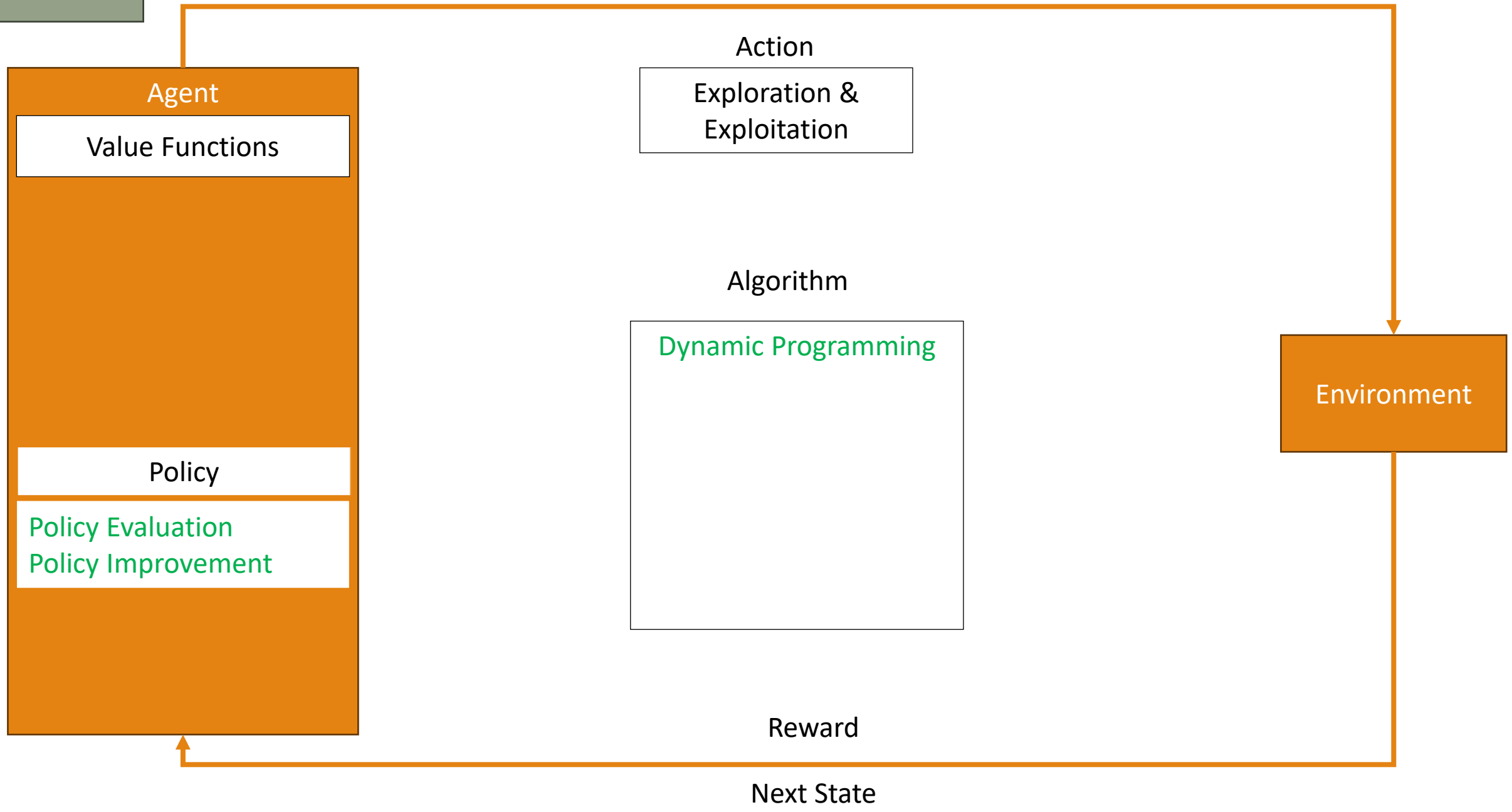
Lecture 4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Policy Improvement
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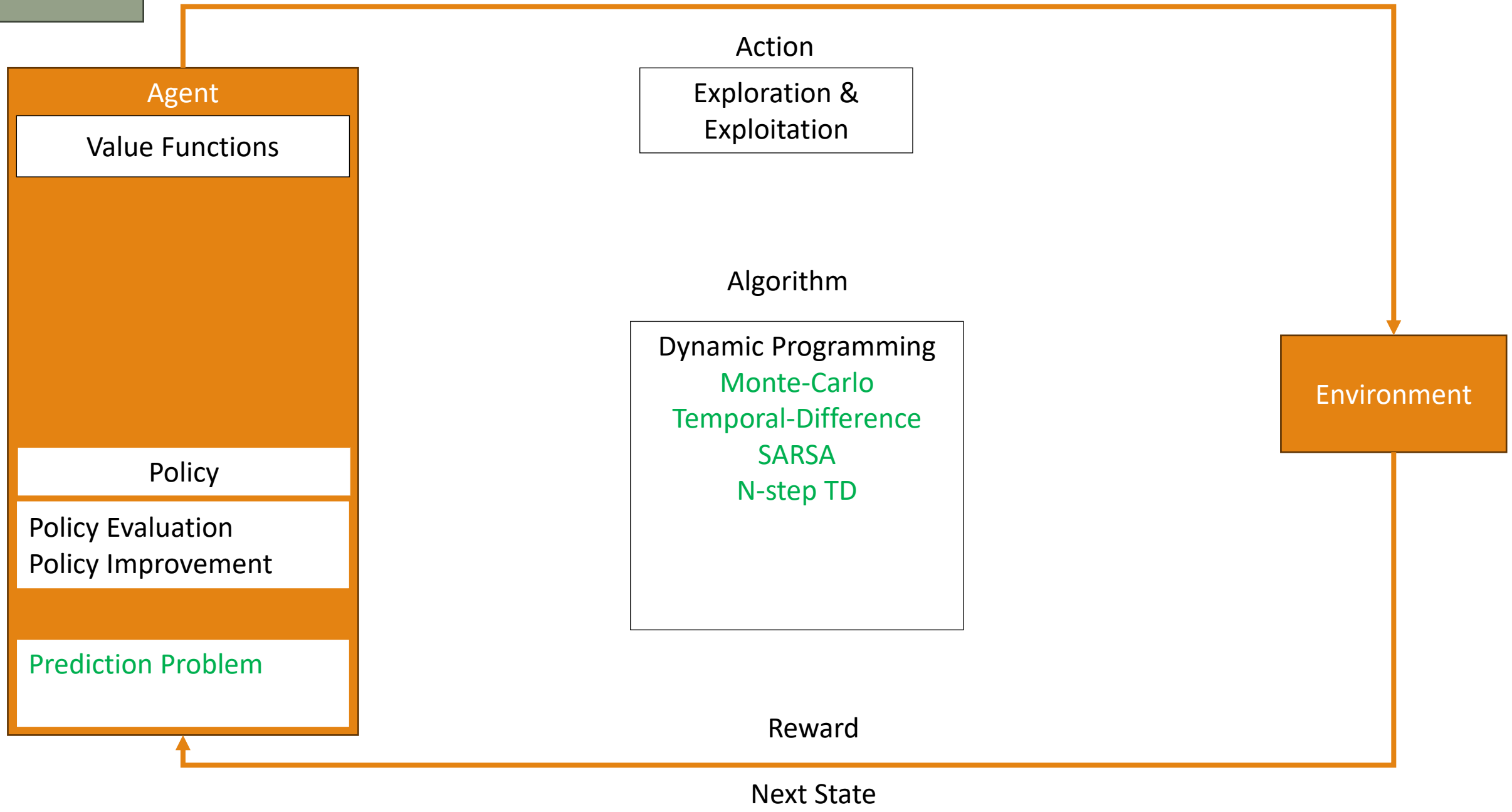
Next State

Lecture 5


Action
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Temporal-Difference
SARSA
N-step TD
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Policy
Policy Evaluation
Policy Improvement
Prediction Problem
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Next State

Four main Bellman Equations

Bellman Expectation Equation

$$v_\pi(s) = \mathbb{E}[R_{t+1} + \gamma v_\pi(S_{t+1}) \,|S_t = s, A_t \sim \pi(s)]$$
$$q_\pi(s,a) = \mathbb{E}[R_{t+1} + \gamma q_\pi(S_{t+1}, A_{t+1}) \,|S_t = s, A_t = a]$$

Bellman Optimality Equation

$$v^*(s) = \max_a \mathbb{E}[R_{t+1} + \gamma v^*(S_{t+1}) \,|S_t = s, A_t = a]$$
$$q^*(s,a) = \mathbb{E}\left[R_{t+1} + \gamma \max_{a'} q^*(S_{t+1}, a') \,\middle|\, S_t = s, A_t = a)\right]$$

Prediction and Control

Prediction

- A problem when we perform **policy evaluation**
- Estimating v_π or q_π

Control

- A problem when we perform **policy optimization**
- Estimating v^* or q^*

Evaluating Policy

$$\pi \geq \pi' \leftrightarrow v_\pi(s) \geq v_{\pi'}(s) \quad , \forall s$$

One policy is better than or equal to another **if and only** if its value function is greater or equal to another.

Policy Evaluation

Given a policy, we want to estimate this

$$v_\pi(s) = \mathbb{E}[R_{t+1} + \gamma v_\pi(S_{t+1}) \,|s, \pi]$$

We initialize all value to zero, and we iterate the equation above as an update

$$v_{k+1}(s) = \mathbb{E}[R_{t+1} + \gamma v_k(S_{t+1}) \,|s, \pi] \quad , \quad \forall s$$

Note that

$$v_\pi(s) = \sum_a \pi(a|s) \sum_r \sum_{s'} p(r, s'|s, a)(r + \gamma v_\pi(s'))$$

If $v_{k+1}(s) = v_k(s), \forall s$, we solve v_π.

Policy Improvement

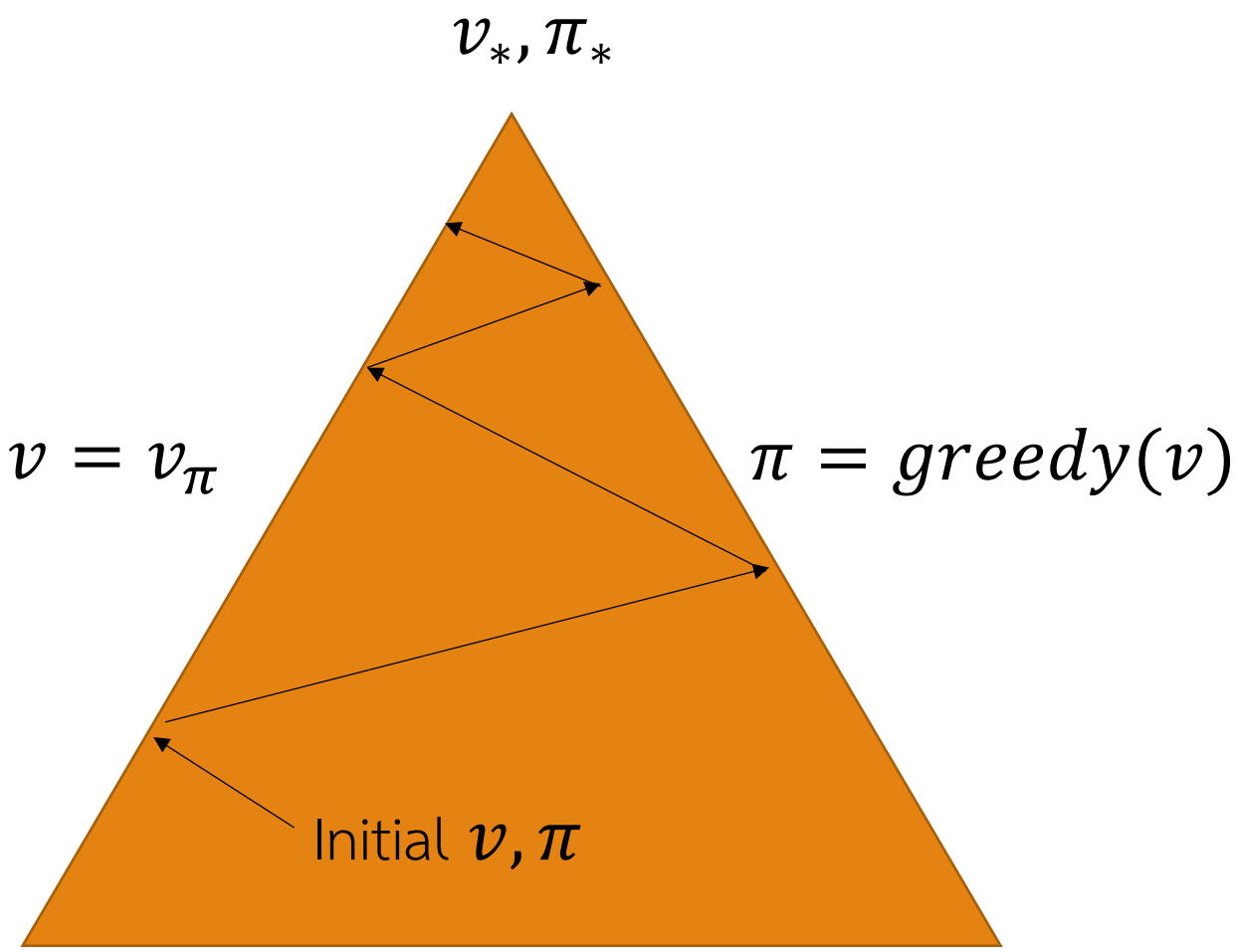
The example shows that we can improve a policy when we learn the value.

- In the example, we do not improve any policy

$$\forall s: \pi_{new}(s) = argmax_a \, q_\pi(s,a)$$
$$= argmax_a \mathbb{E}[R_{t+1} + \gamma v_\pi(S_{t+1}) | S_t = s, A_t = a]$$

We use the new policy to evaluate, and repeat.

Policy Iteration



Starting with initial v, π

Loop do:

Perform policy evaluation,

$$v = v_\pi$$

Perform policy improvement,

$$\pi' \geq \pi$$

Value Iteration

Or we can learn the policy on the fly

- We update the policy every iteration

We can take the Bellman optimality equation as an update

$$v_{k+1}(s) \leftarrow \max_{a} \mathbb{E}[R_{t+1} + \gamma v_k(S_{t+1}) \,|S_t = s, A_t = a] \quad , \quad \forall s$$

It is the same as **policy iteration** with k = 1. Basically, we improve the policy every iteration.

Model-Free Learning

Dynamic Model

$$v_\pi(s) = \sum_a \pi(a|s) \sum_r \sum_{s'} p(r, s'|s, a)(r + \gamma v_\pi(s'))$$

The value function above contains the dynamic of the problem

- $p(r, s'|s, a)$ tells the transition model of the problem. This is essentially the dynamic of the problem.

Key Question: What if we do not know that? Can the agent still learn?

- Spoiler Alerted: Yes, it can. (Otherwise, we wouldn't have this class)

Our Plan

Last Lecture

- We solved a **known MDP** by dynamic programming

This Lecture

- **Model-free prediction** to **estimate values** in an **unknown MDP**
- A brief touch of Deep Reinforcement Learning

Next Lecture

- **Model-free control** to **optimize values** in an **unknown MDP**

Monte Carlo Algorithm

Monte Carlo Algorithm

To learn without a model, we can **sample the experience**

- Let's just act until the end and learn from that

Monte Carlo is a sampling method

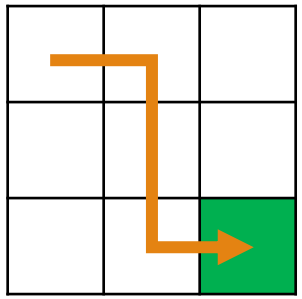
- Direct sampling of episodes
- Model-free
- No knowledge of MDP required

Monte Carlo Example

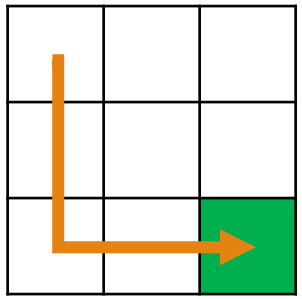
To sample the experience, given a policy π, the agent just plays out until termination.

For example, given a grid world and a robot that can move up, down, left, right, the samples are:

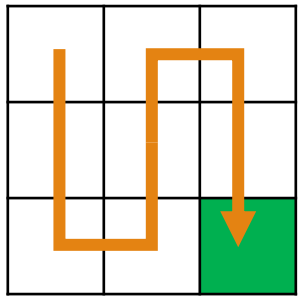
Episode 1



Episode 2



Episode 3



Monte-Carlo Policy Evaluation

Given the sequential decision problem, MC learns v_π from episodes under policy π

$$S_1, A_1, R_2, \dots, S_k \sim \pi$$

The return is calculated given an ending time T

$$G_t = R_{t+1} + \gamma R_{t+2} + \cdots + \gamma^{T-t-1} R_T$$

The value function is then calculated from the expected return

$$v_\pi(s) = \mathbb{E}[G_t | S_t = s, \pi]$$

We can use **sample average return** instead of **expected return**

- This is called **Monte Carlo policy evaluation**

Part 1 Summary: Monte Carlo

Input: policy π, $num\_episodes$
Initialize $N(s,a) = 0$ for all $s \in \mathcal{S}, a \in \mathcal{A}(s)$
Initialize $returns\_sum(s,a) = 0$ for all $s \in \mathcal{S}, a \in \mathcal{A}(s)$
for $i \leftarrow 1$ ***to*** $num\_episodes$ **do**
  Generate an episode $S_0, A_0, R_1, \ldots, S_T$ using π
  for $t \leftarrow 0$ ***to*** $T-1$ **do**
    if (S_t, A_t) *is a first visit (with return* G_t*)* **then**
      $N(S_t, A_t) \leftarrow N(S_t, A_t) + 1$
      $returns\_sum(S_t, A_t) \leftarrow returns\_sum(S_t, A_t) + G_t$
  end
end
$Q(s,a) \leftarrow returns\_sum(s,a)/N(s,a)$ for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$
return Q

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Blackjack Example

200 States

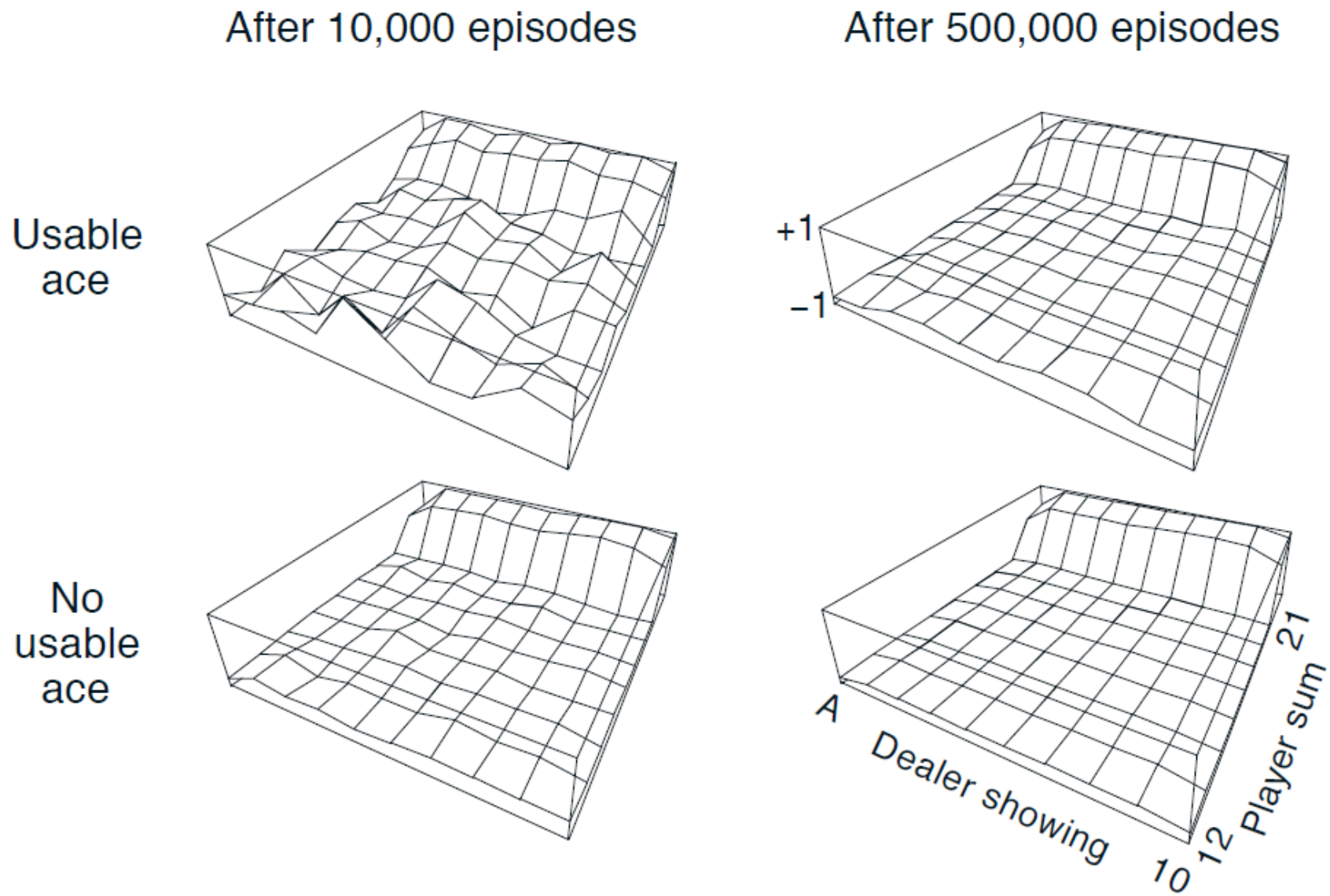
- Current hand (12-21)
- Dealer’s hand (A-10)
- Our usable ace (yes, no)

Actions: stick, draw

Reward

- Stick: +1 if win, 0 if draw, -1 if lose
- Draw: -1 if lose, 0 otherwise

Transitions: draw if sum < 12


After 10,000 episodes
After 500,000 episodes
Usable ace
No usable ace
+1
−1
A
Dealer showing
10
12
Player sum
21

Function Approximation

Value Function Approximation

In dynamic programming with MDP, we use **lookup tables** to learn value function.

- A mapping of state s to a value $v(s)$
- A mapping of state-action pair s, a to a state-action value $q(s, a)$

But we cannot do that for a large MDP

- Infeasible to store all information due to the **curse of dimensionality**
- **Too slow to learn** the value of each state **individually**
- States are often **not fully observable**

Value Function Approximation

For large MDP, we approximate the value function instead

$$v_{\mathbf{w}}(s) \approx v_{\pi}(s)$$
$$q_{\mathbf{w}}(s,a) \approx q_{\pi}(s,a)$$

The agent will learn the value and update (learnable) parameter $\mathbf{w}$.

- Monte Carlo Algorithm
- Temporal Difference Learning

Agent State Update

For large MDP, if the environment state is not fully observable, we can update the agent state by

$$S_t = u_\omega(S_{t-1}, A_{t-1}, O_t)$$

with parameters $\omega \in \mathbb{R}^n$

Linear Value Function Approximation

Let's start with a simple, useful, and special case: a linear function

A state is represented by a **feature vector**

$$\mathbf{x}(s) = \begin{bmatrix} x_1(s) \\ \vdots \\ x_m(s) \end{bmatrix}$$

$\mathbf{x} : S \Rightarrow \mathbb{R}^m$ is a mapping from an agent state to features

Linear Value Function Approximation

One way to approximate a value function is to find a linear combination of features.

$$v_{\mathbf{w}}(s) = \mathbf{w}^{\mathrm{T}}\mathbf{x}(s) = \sum_{j=1}^{n} w_j x_j(s)$$

To learn the function, we can find an objective loss.

$$L(\mathbf{w}) = \mathbb{E}_{S\sim d}\left[\left(v_{\pi}(S) - \mathbf{w}^{\mathrm{T}}\mathbf{x}(s)\right)^2\right]$$

Linear Value Function Approximation

$$L(\mathbf{w}) = \mathbb{E}_{S\sim d}\left[\left(v_\pi(S) - \mathbf{w}^{\mathrm{T}}\mathbf{x}(s)\right)^2\right]$$

Then we can use stochastic gradient descent to update the parameter

- Stochastic gradient descent is a gradient descent algorithm that calculates loss of 1 input at a time
- Since we sample the problem, we can update the parameter using SGD

$$\nabla_{\mathbf{w}} v_{\mathbf{w}}(S_t) = \mathbf{x}(S_t) = \mathbf{x}_t \qquad \Rightarrow \qquad \Delta\mathbf{w} = \alpha\left(v_\pi(S_t) - v_{\mathbf{w}}(S_t)\right)\mathbf{x}_t$$

Update = step-size x prediction error x feature vector

Function Approximation Example

Let's consider a bandit problem. If we want to learn the value function, we can find it through the objective loss function.

- q is a parametric function with parameters $\mathbf{w}$. For example, q could be a neural network.

$$L(\mathbf{w}) = \frac{1}{2}\mathbb{E}[(R_{t+1} - q_{\mathbf{w}}(S_t, A_t))^2]$$

Then we update the weight by:

$$\begin{aligned}\mathbf{w}_{t+1} &= \mathbf{w}_t - \alpha\nabla_{\mathbf{w}_t} L(\mathbf{w}_t) \\ &= \mathbf{w}_t - \alpha\nabla_{\mathbf{w}_t}\frac{1}{2}\mathbb{E}[(R_{t+1} - q_{\mathbf{w}}(S_t, A_t))^2] \\ &= \mathbf{w}_t + \alpha\mathbb{E}\left[(R_{t+1} - q_{\mathbf{w}}(S_t, A_t))\nabla_{\mathbf{w}_t} q_{\mathbf{w}}(S_t, A_t)\right]\end{aligned}$$

Function Approximation Example

$$\mathbf{w}_{t+1} = \mathbf{w}_t + \alpha \mathbb{E}\left[(R_{t+1} - q_{\mathbf{w}}(S_t, A_t))\nabla_{\mathbf{w}_t} q_{\mathbf{w}}(S_t, A_t)\right]$$

If we use a linear function approximation,

$$q(s,a) = \mathbf{w}^{\mathrm{T}}\mathbf{x}(s,a)$$

The gradient becomes

$$\nabla_{\mathbf{w}_t} q_{\mathbf{w}}(S_t, A_t) = \mathbf{x}(s,a)$$

Then the SGD update is

$$\mathbf{w}_{t+1} = \mathbf{w}_t + \alpha(R_{t+1} - q_{\mathbf{w}}(S_t, A_t))\mathbf{x}(s,a)$$

Linear Update = step-size x prediction error x feature vector

Non-linear Update = step-size x prediction error x feature vector

Monte Carlo

MC algorithm can be used to learn value function

- However, the algorithm must wait until the end of an episode to learn something
- Return can have high variance

Alternatives?

- Yes! Temporal Difference Learning

Temporal Difference Learning

Temporal Difference Learning

Given a Bellman equation,

$$v_\pi(s) = \mathbb{E}[R_{t+1} + \gamma v_\pi(S_{t+1}) \,|S_t = s, A_t \sim \pi(S_t)]$$

We can update values by iterating using the following update function,

$$v_{k+1}(s) = \mathbb{E}[R_{t+1} + \gamma v_k(S_{t+1}) \,|S_t = s, A_t \sim \pi(S_t)]$$

Temporal Difference Learning

$$v_{k+1}(s) = \mathbb{E}[R_{t+1} + \gamma v_k(S_{t+1}) \,|S_t = s, A_t \sim \pi(S_t)]$$

Instead of calculating the expected return, we can sample the value.

$$v_{t+1}(S_t) = R_{t+1} + \gamma v_t(S_{t+1})$$

But instead of updating everything on the noisy value, we can update the value a little bit instead.

$$v_{t+1}(S_t) = v_t(S_t) + \alpha_t(R_{t+1} + \gamma v_t(S_{t+1}) - v_t(S_t))$$

MC Prediction vs TD Prediction

Prediction problem: learn v_π online from experience under policy π

Monte Carlo:

- Update value $v_n(S_t)$ with respect to sample return G_t

$$v_n(S_t) = v_n(S_t) + \alpha(G_t - v_n(S_t))$$

Temporal-difference learning:

- Update value $v_t(S_t)$ with respect to estimated return $R_{t+1} + \gamma v(S_{t+1})$

$$v_{t+1}(S_t) = v_t(S_t) + \alpha_t(R_{t+1} + \gamma v_t(S_{t+1}) - v_t(S_t))$$

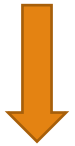
Temporal Difference Learning

$$v_{t+1}(S_t) = v_t(S_t) + \alpha(R_{t+1} + \gamma v_t(S_{t+1}) - v_t(S_t))$$

Target

Temporal Difference Learning

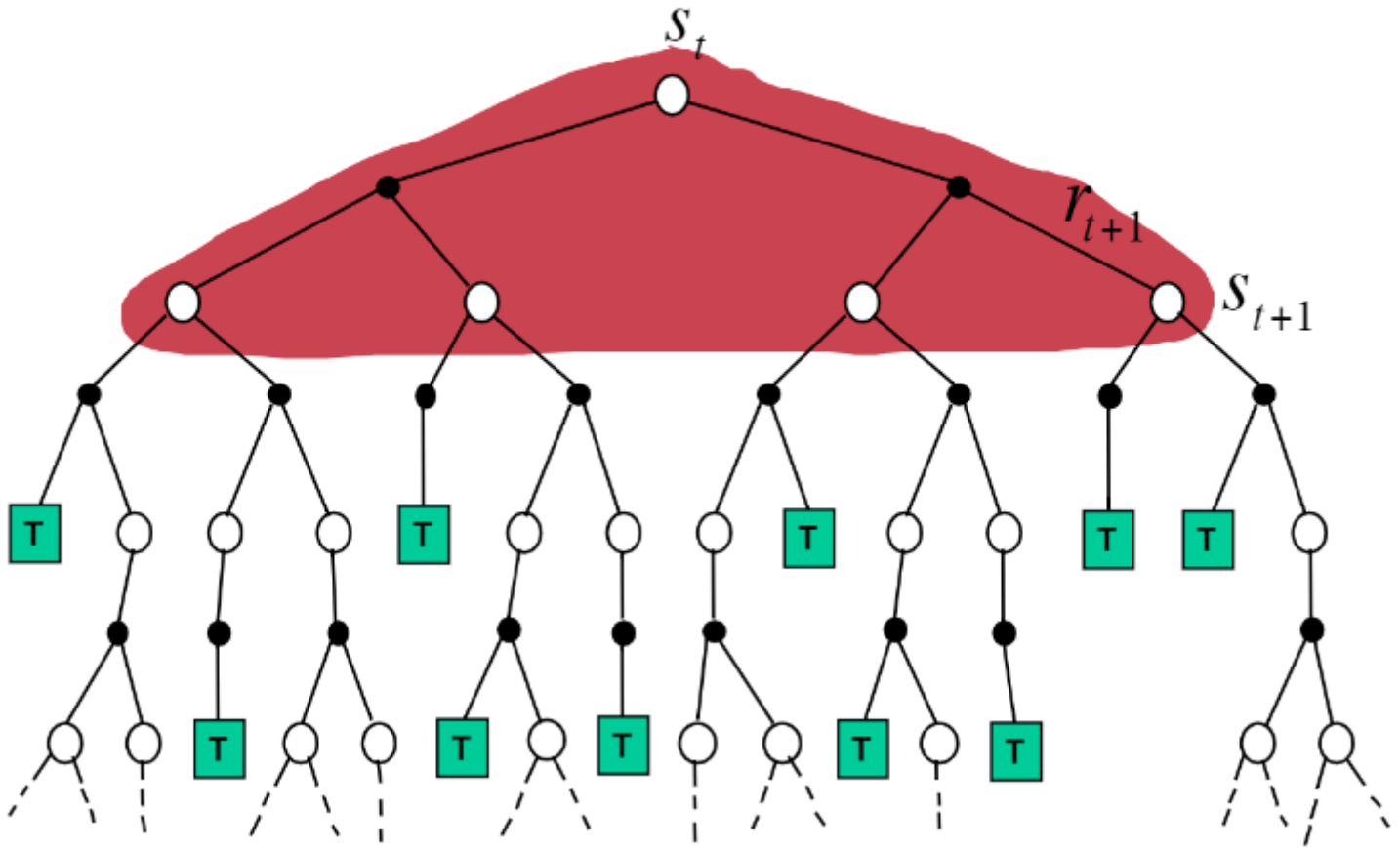
$$v_{t+1}(S_t) = v_t(S_t) + \alpha(R_{t+1} + \gamma v_t(S_{t+1}) - v_t(S_t))$$



TD Error

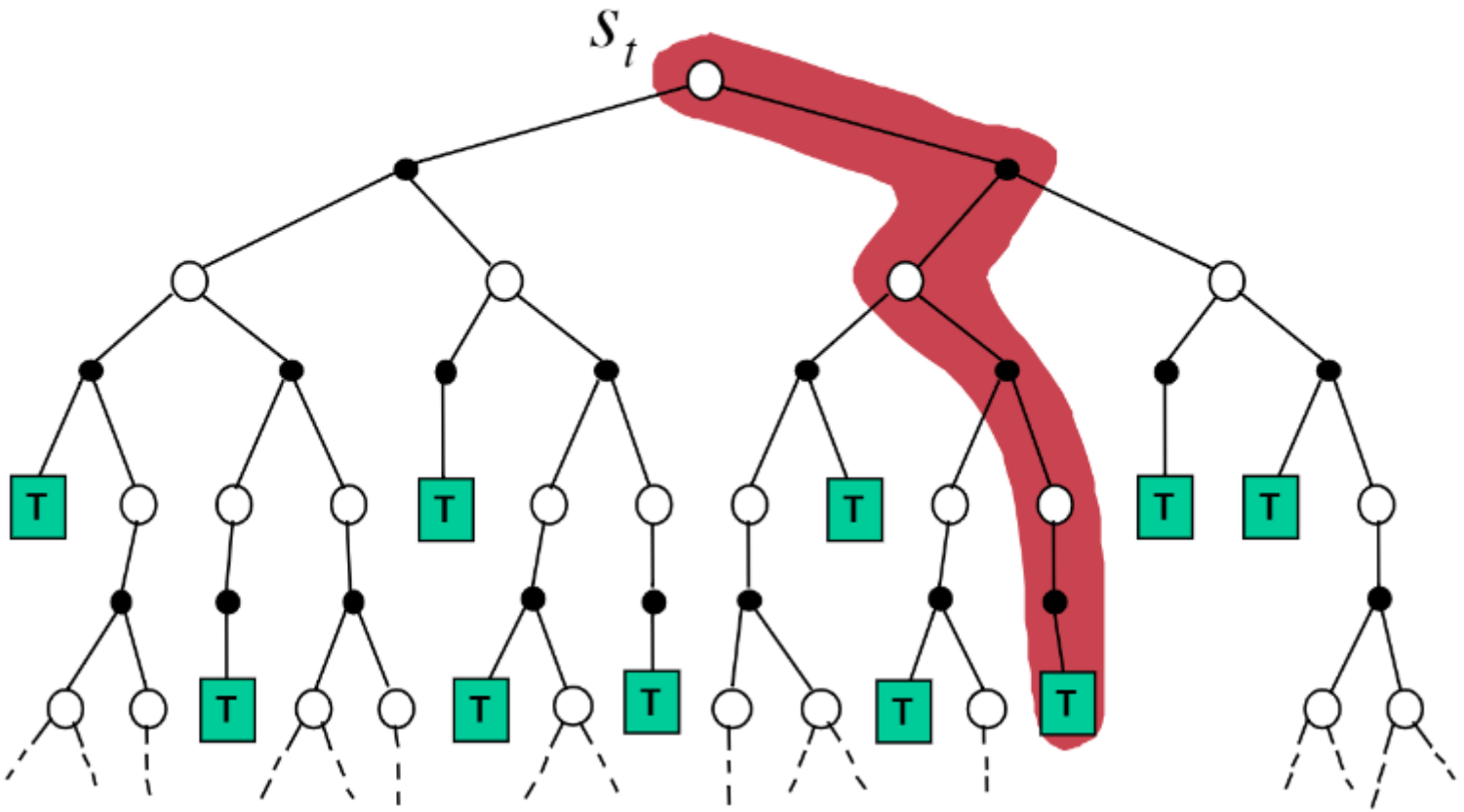
Dynamic Programming Backup

$$v(S_t) \leftarrow \mathbb{E}\left[R_{t+1} + \gamma v(S_{t+1}) \mid A_t \sim \pi(S_t)\right]$$



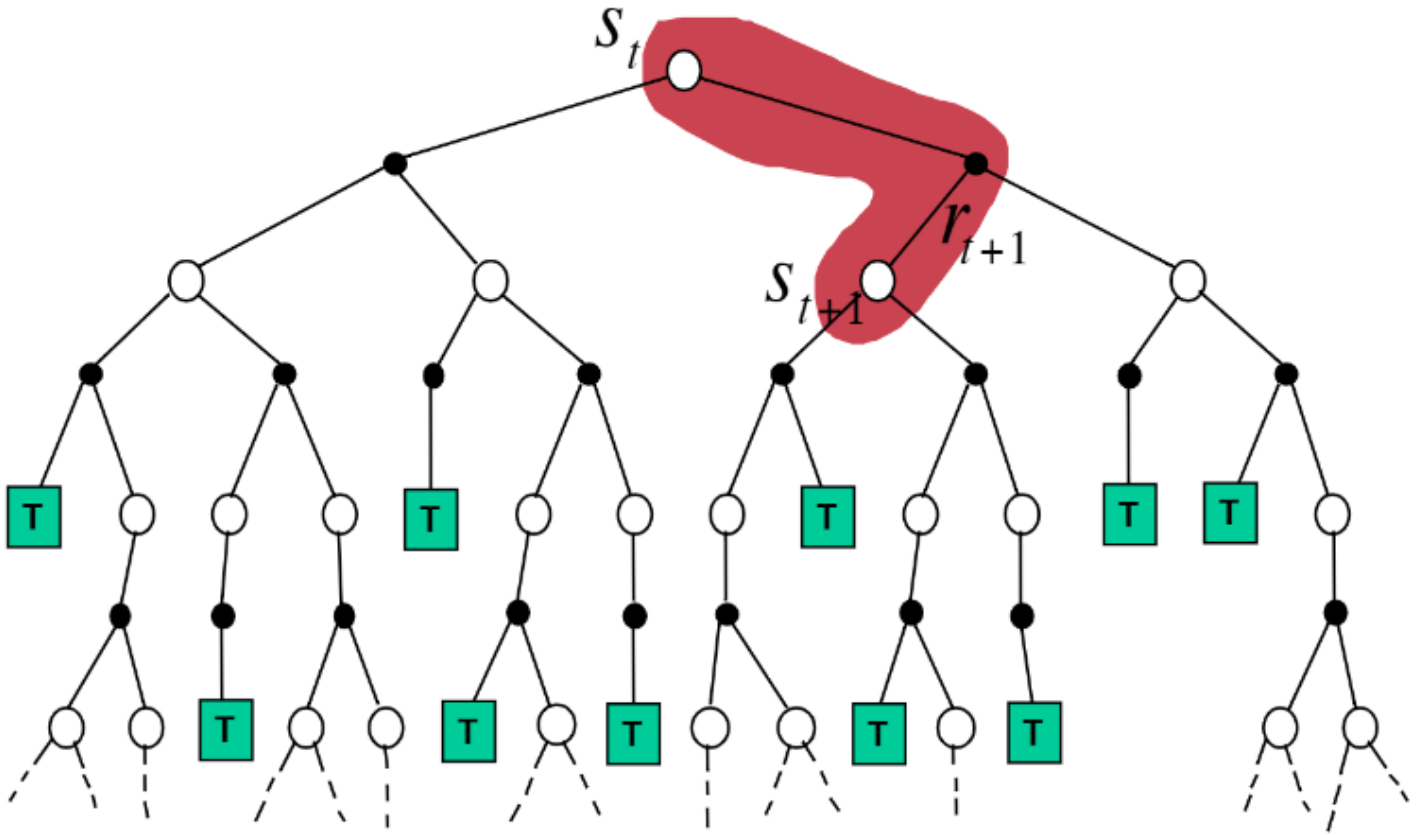
Monte-Carlo Backup

$$v(S_t) \leftarrow v(S_t) + \alpha\,(G_t - v(S_t))$$



Temporal-Difference Backup

$$v(S_t) \leftarrow v(S_t) + \alpha \left(R_{t+1} + \gamma v(S_{t+1}) - v(S_t)\right)$$



Bootstrapping and Sampling

Bootstrapping

- Use the estimate of the next state to update
- DP and TD use
- MC does not

Sampling

- Update samples an expectation
- MC and TD sample
- DP does not

Temporal Difference Learning

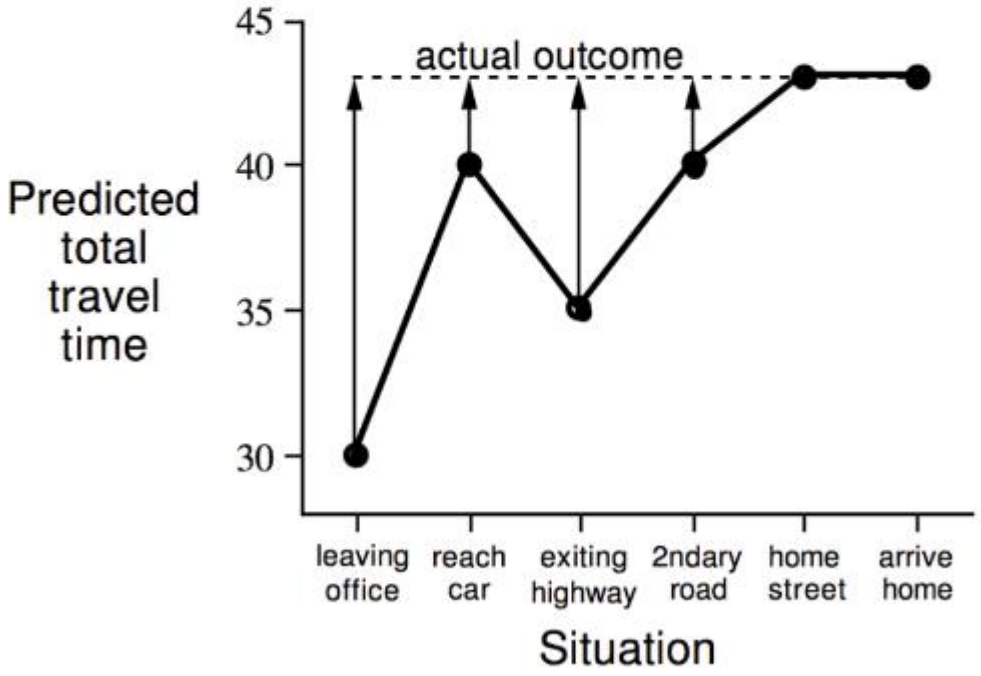
We can also learn action values by updating value $q_t(S_t, A_t)$ with respect to estimated return $R_{t+1} + \gamma v(S_{t+1}, A_{t+1})$

$$q_{t+1}(S_t, A_t) = q_t(S_t, A_t) + \alpha_t(R_{t+1} + \gamma q_t(S_{t+1}, A_{t+1}) - q_t(S_t, A_t))$$

This algorithm is called SARSA, because it uses $S_t, A_t, R_{t+1}, S_{t+1}, A_{t+1}$

State	Elapsed Time (minutes)	Predicted Time to Go	Predicted Total Time
leaving office	0	30	30
reach car, raining	5	35	40
exit highway	20	15	35
behind truck	30	10	40
home street	40	3	43
arrive home	43	0	43

Changes recommended by Monte Carlo methods (α=1)



Changes recommended by TD methods (α=1)

TD vs MC

TD can learn **before** knowing the final outcome

- TD learns online every step.
- MC must wait until the end of the episode.

TD can learn **without** the final outcome

- TD can learn even if the full interaction sequence is incomplete. MC must use the complete sequence.
- TD can be used in a continuing environment. MC environment must terminate.

Bias-Variance Trade-off

Given MC return,

$$G_t = R_{t+1} + \gamma R_{t+2} + \cdots + \gamma^{T-t-1} R_T$$

This is an **unbiased** estimation of $v_\pi(S_t)$

Given TD target,

$$R_{t+1} + \gamma v_t(S_{t+1})$$

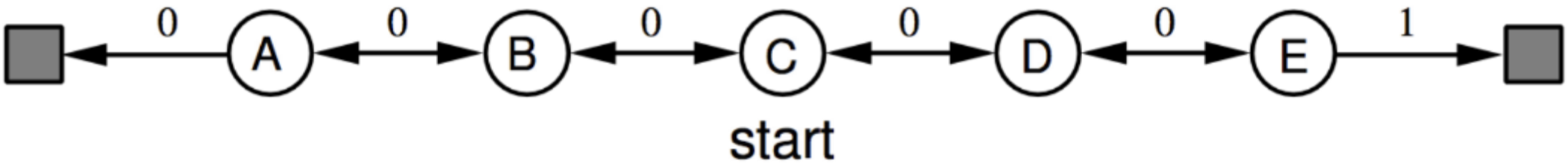
This is a **biased** estimation of $v_\pi(S_t)$

Bias-Variance Trade-off

However, TD has **lower variance**

- MC return estimates from many random actions, transitions, and rewards
- TD target estimates from one random action, transition, and reward
- Variation of action, transition, and reward makes the TD target have less variance

Random Walk Example

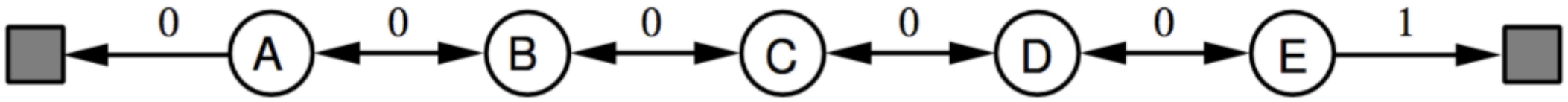


This game has 7 states, with uniform random transitions (50% left, 50% right)

We can see that the true values of all states are:

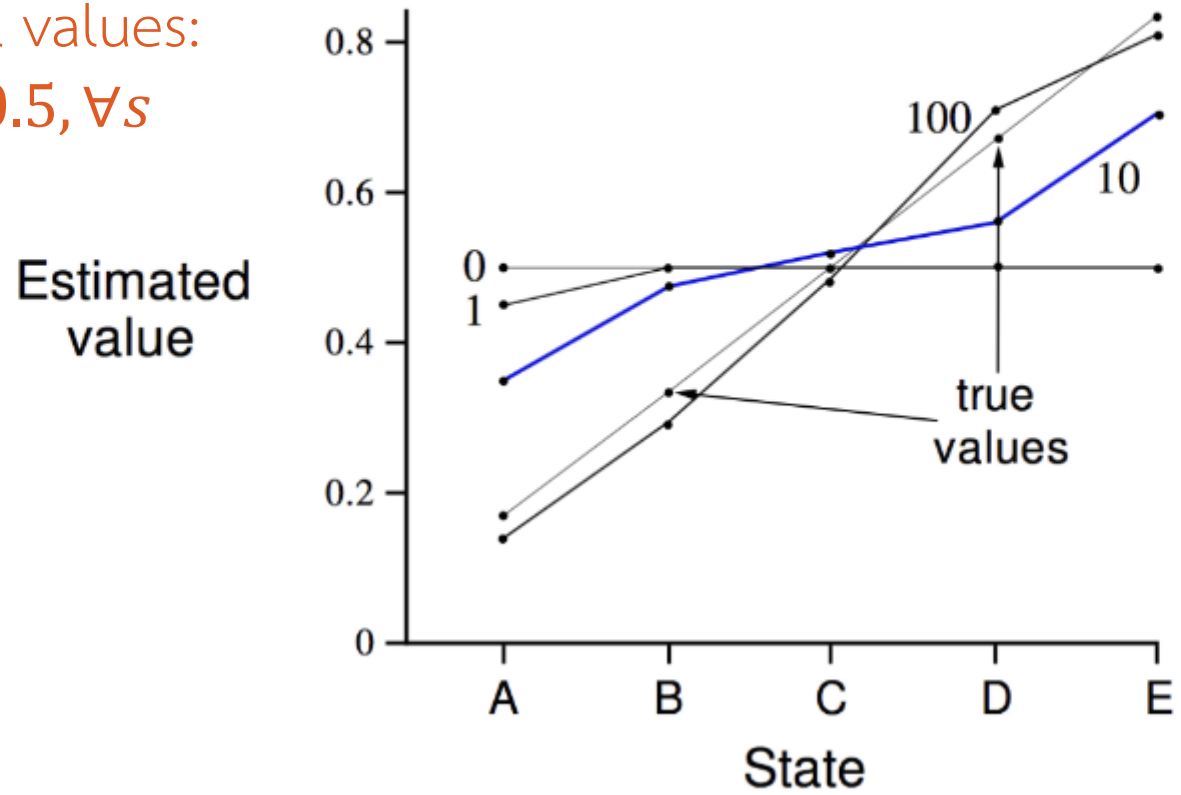
$$v(A) = \frac{1}{6}, v(B) = \frac{2}{6}, v(C) = \frac{3}{6}, v(D) = \frac{4}{6}, v(E) = \frac{5}{6}$$

Random Walk Example

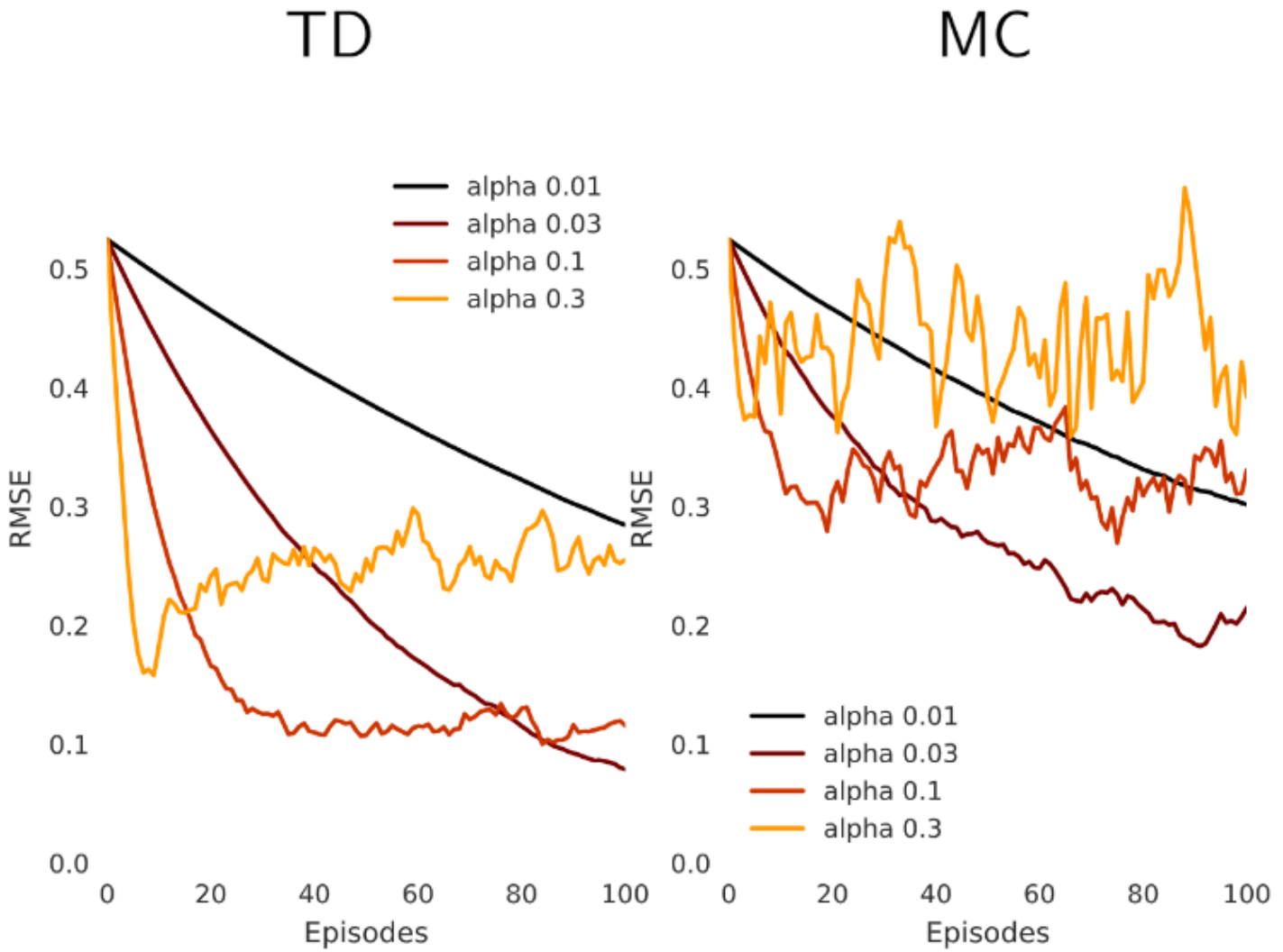


Given that initial values:

$$v(s) = 0.5, \forall s$$



Random Walk Example



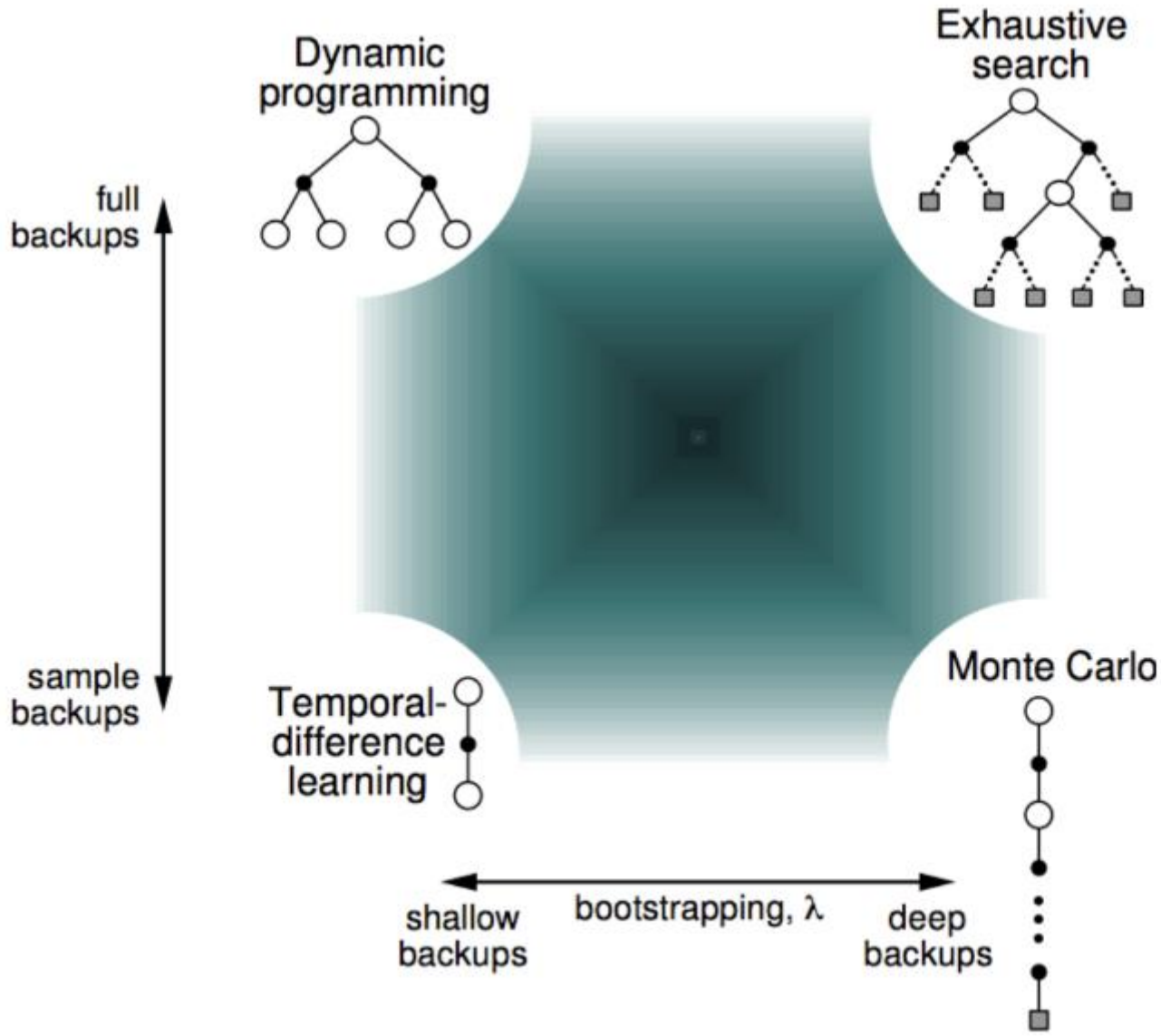
TD vs MC

TD exploits Markov property

- TD can help in fully-observable environments

MC does not exploit Markov Property

- MC can help in partially-observable environments


Dynamic programming
Exhaustive search
full backups
sample backups
Temporal-difference learning
Monte Carlo
shallow backups
bootstrapping, λ
deep backups

Multi-Step Updates

Multi-Step Updates

TD and MC tackle the problem from different angles

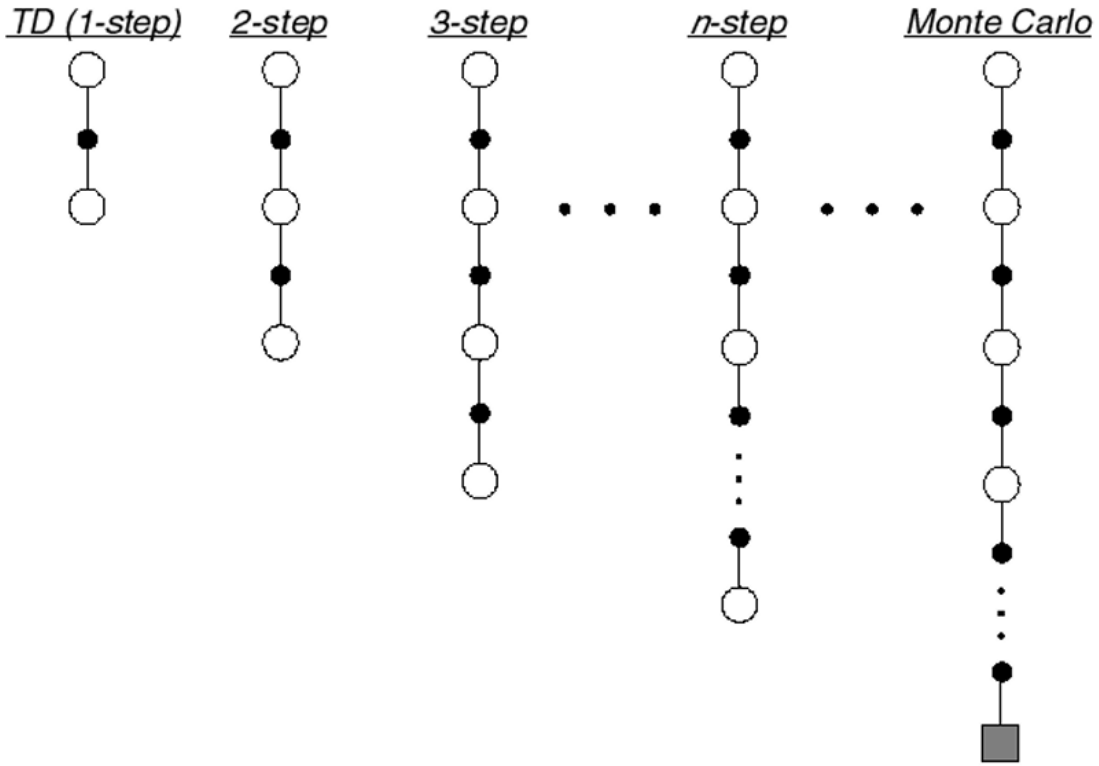
- TD estimates, and bootstrap
- MC use true return, but is noisy

Can we do something between?

- Multi-Step Prediction

Multi-Step Prediction

We can modify TD target to look n steps into the future



Multi-Step Returns

We can arrange TD and MC as:

TD $n = 1$ $G_t^{(1)} = R_{t+1} + \gamma v(S_{t+1})$

$n = 2$ $G_t^{(2)} = R_{t+1} + \gamma R_{t+2} + \gamma^2 v(S_{t+2})$

$\vdots$

MC $n = \infty$ $G_t^{(\infty)} = R_{t+1} + \gamma R_{t+2} + \ \dots \ + \gamma^{T-t-1} R_T$

In general, we can formalize n-step return as

$$G_t^{(n)} = R_{t+1} + \gamma R_{t+2} + \ \dots \ + \gamma^{n-1} v(S_{t+1})$$

Multi-Step Returns

In general, we can formalize n-step return as

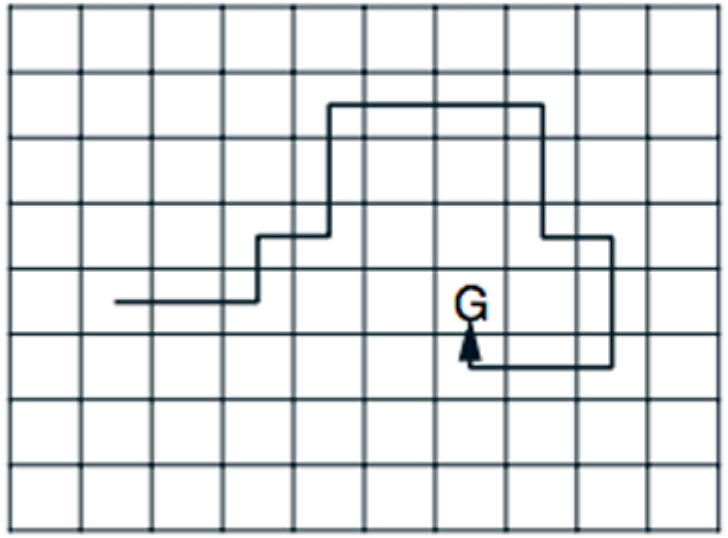
$$G_t^{(n)} = R_{t+1} + \gamma R_{t+2} + \ \ldots \ + \gamma^{n-1} v(S_{t+1})$$

We can define multi-step temporal-difference learning as

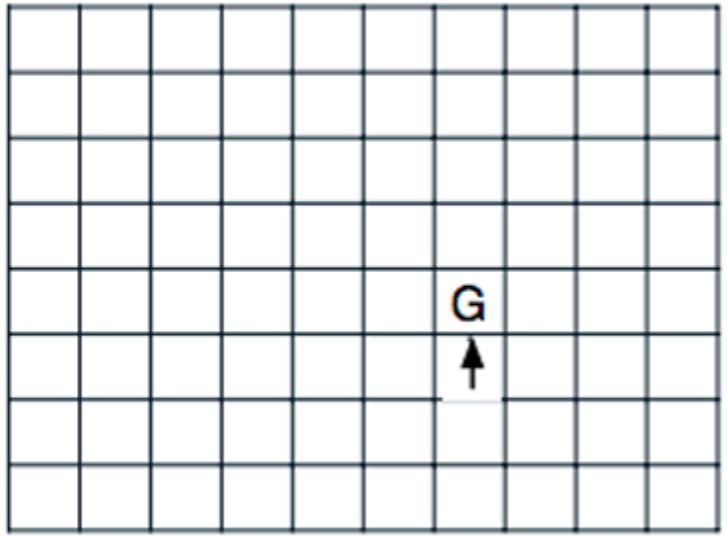
$$v(S_t) \leftarrow v(S_t) + \alpha(G_t^{(n)} - v(S_t))$$

Example

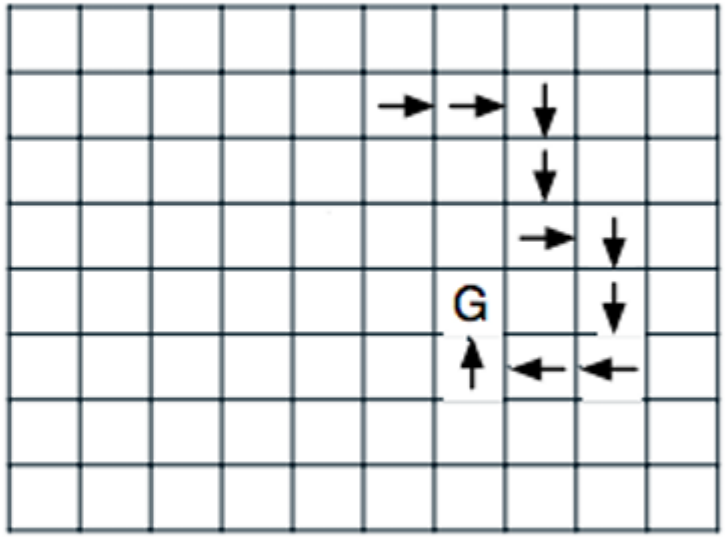
Path taken



Action values increased by one-step Sarsa

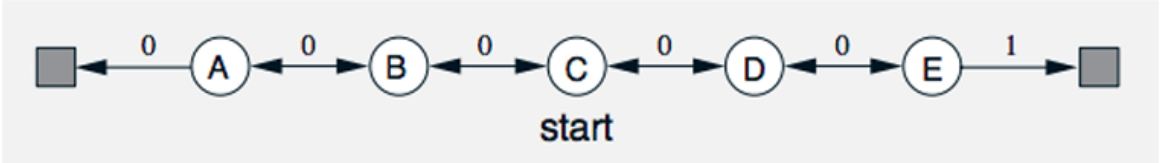


Action values increased by 10-step Sarsa

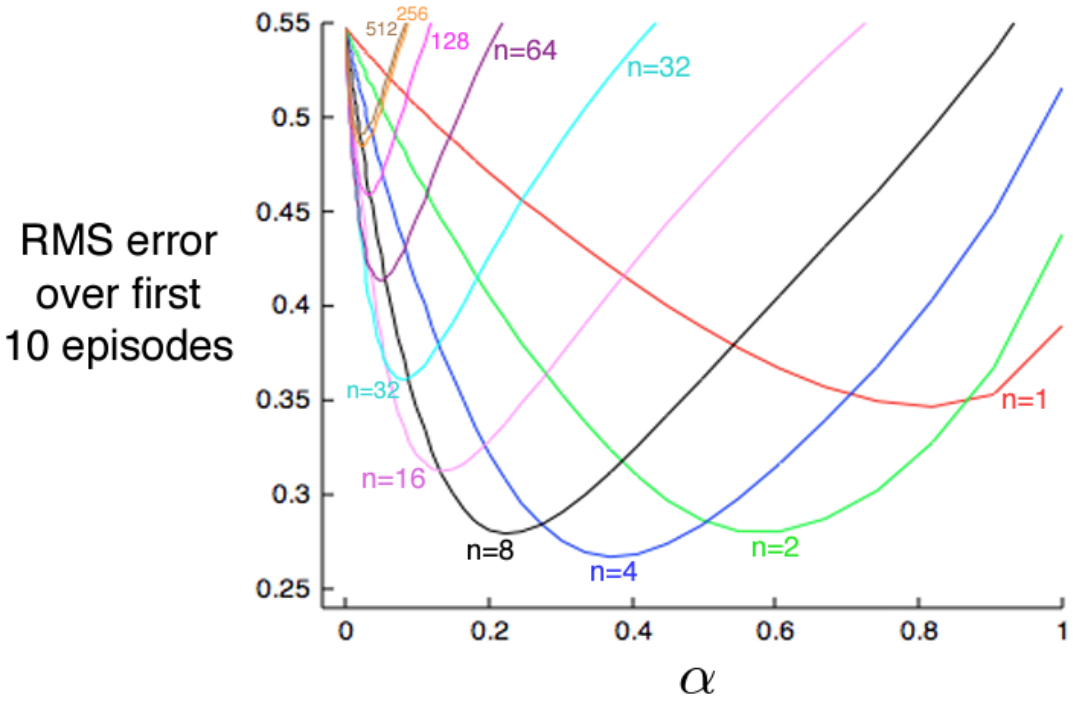


(Reminder: SARSA is TD for action values $q(s, a)$)

Example



..., but with 19 states, rather than 5



Mixing Multi-Step Returns

Consider a multi-step return equation,

$$G_t^{(n)} = R_{t+1} + \gamma R_{t+2} + \ \dots \ + \gamma^{n-1} v(S_{t+1})$$

Multi-step returns bootstrap on one state, $v(S_{t+n})$. However, we can modify the equation so you can bootstrap a little bit on many states

$$G_t^{\lambda} = R_{t+1} + \gamma((1-\lambda)v(S_{t+1}) \ + \lambda G_{t+1}^{\lambda})$$

Multi-Step Returns

$$G_t^{\lambda} = R_{t+1} + \gamma((1-\lambda)v(S_{t+1}) \ + \lambda G_{t+1}^{\lambda})$$

If we consider:

$\lambda = 0$ $\quad G_t^{\lambda=0} = R_{t+1} + \gamma v(S_{t+1})$ (TD)

$\lambda = 1$ $\quad G_t^{\lambda=1} = R_{t+1} + \gamma G_{t+1}$ (MC)

Multi-Step Returns

Using multi-step returns,

- We can utilize benefits of TD and MC
- Bootstrapping may cause bias
- Monte Carlo involves high variance

Generally, using multi-step returns are good